MATH1090/1098
(Honours) Introduction to Set Theory

The Chinese University of Hong Kong

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Chapter 0

Advice to the student

- Sections marked $\star$ are optional for MATH1090 students.
- Bring a paper and pencil / pens to class, and leave your phone / laptop / tablet in your bag.
- During the in-class exercises, work with the people next to you. Ask lots of questions! You will learn better if you say your thoughts out loud.
- This class will have graded assignments and optional exercises. However, most of your grade comes from the midterm and final marks. The assignments are mainly there to help you practice. We suggest you spend 10 minutes every day doing a few optional exercises; this is the best way to internalize the subject.

Chapter 1

Logic

1.1 The general shape of a mathematical argument

This first chapter is about the different kind of arguments used in mathematics. Very roughly, these look like

$$\begin{array}{c} \text{Hypothesis A} \\ \text{Hypothesis B} \\ \hline \text{Conclusion C} \end{array}$$

or

$$\begin{array}{c} A \\ B \\ \hline \therefore C \end{array}$$

or in plain English: 'If A and B are true, then C is also true'. More concisely: 'C follows from A and B'.

We want to know:

Question 1. *What kind of statements can we use as hypotheses or conclusions?*

Question 2. *How do we deduce the conclusion from the hypotheses?*

Of course, in life there is no single correct answer to these questions. In mathematics, however, there is a more or less universal answer, which you will learn in the next few sections.

Propositions The simplest type of statement we will consider is a *proposition*, often denoted by a letter like A or B.

A proposition is a sentence or claim which can be either true or false. For example,

A: The moon is made of cheese.

and

B: I want new socks.

are propositions.[1] The sentence

Oh NOOOOO not again!!!

is not a proposition, because it is not a true/false claim.

1.2 Propositional logic.

Suppose you are investigating a murder. A, B, C, D, E, F, G, H represent your hypotheses, each of which can be either true or false; they are thus examples of propositions.

- A: the murder happened during the night.
- B: the murder happened during the day.
- C: Omar was asleep during the night.
- D: Michael was asleep during the day.
- E: Mykola was asleep between 3am and 11am.
- F: Omar is the murderer.
- G: Michael is the murderer.
- H: Mykola is the murderer.

These hypotheses are not independent. For instance:

1. If A is false, then B is true.
2. If B and D are true, then G is false.

We would like a good way to think about these dependencies, and to build complex statements out of them. Our tool will be *propositional logic*, also called *zero-th order* or *Boolean logic*.

[1] A good test: check if you can add 'It is true that...' or 'It is not true that...' to the beginning of the sentence.

Propositions as variables Let us be a little more formal. In propositional logic, we treat each proposition like a variable which can be either *true (T)* or *false (F)*. The *truth value* of A is T if A is true, and F if A is false.

We thus forget all the irrelevant details of a proposition like A or D: *Who was murdered? Who is Michael? Why was he sleeping during the day?*. All we care about is whether A or D are TRUE or FALSE.

We write $A = T$ to indicate 'A is true' and $A = F$ to indicate 'A is false'. We call this the *truth value*, or simply the *value*, of the proposition.

Remark 1. *We will think of propositions as analogous to variables like a, b, c in algebraic equations like $a + b = b \cdot c$. Instead of taking numerical values like* 5 *or* 1.23*, propositions take only the values T or F, and are thus "simpler". We will discuss the analogues of $+$, $\cdot$, and $=$ below.*

1.3 The alphabet of propositional logic

Propositional logic is a kind of language, a bit like English or Chinese. It uses a very small alphabet of symbols. Here they are, labeled by their English names.

- $\wedge$ (AND)
- $\vee$ (OR)
- $\neg$ (NOT)
- $\leftrightarrow$ (IF AND ONLY IF or IS EQUIVALENT TO)
- $\rightarrow$ (IMPLIES)

A *Boolean formula* is a formula constructed using our propositions $A, B, C, ..., H$ together with the above symbols. For example, we can write

If A is false, then B is true

as the Boolean formula

$$\neg A \rightarrow B. \tag{1.1}$$

We can write

If B and D are true, then G is false

as

$$(B \wedge D) \rightarrow \neg G. \tag{1.2}$$

Exercise 1. *Can you write down some more logical dependencies among A, B, C, D, E, F, G, H as Boolean formulas?*

In our analogy, the Boolean operations like $\neg, \vee$ are like the symbols $+$ and $\cdot$ which we can use to take variables and form more complex algebraic formulas out of.

Boolean grammar We do not allow arbitrary strings of Boolean symbols when making Boolean formulas. For example,

$$A \vee \vee \vee \neg$$

is not a Boolean formula: it is just nonsense. There is a list of rules for making well-formed Boolean formulas, a bit like the grammar rules of a language. However, we will not spell out explicitly these rules explicitly. Instead, we trust that you will learn to do this intuitively through practice. If you find this confusing, ask us for some more examples.

Order of operations. Consider the formula

$$A \wedge B \vee C.$$

This formula is ambiguous! Does it mean 'Either A and B are both true, or C is true'? Or does does it mean 'A is true and either B or C' is true'?

To make it unambiguous, we add parenthesis. We write

$$(A \wedge B) \vee C$$

when we mean 'Either A and B are both true, or C is true', and we write

$$A \wedge (B \vee C)$$

when we mean 'A is true and either B or C' is true'. Now consider the formula

$$\neg A \wedge B.$$

Does it mean 'A is false and B is true' or 'It is not true that both A and B are true'? Here there is no ambiguity: by convention, we first apply $\neg$, then $\wedge$, so that the first reading is correct.

More generally, the conventional order in which to read a Boolean formula is: first $\neg$, then $\wedge$ and $\vee$, and finally $\rightarrow$ and $\leftrightarrow$. Any remaining ambiguities must be resolved by parentheses.

Different uses of the word *'proposition'* In most of mathematics, the word 'proposition' is usually reserved for a mathematical true/false claim that the author believes to be true (and has a proof of). This is different from its use within propositional logic. In practice, one can usually easily tell from context which meaning is being used.

1.4 Truth tables

Recall that each proposition $A, B, C, ..$ can have one of two values: true or false.

Boolean formulas like

$$\neg A, \quad A \wedge B \quad A \vee B$$

can also be either true or false, depending on whether A and B are true or false.

We can describe the conditions under which they are true or false using 'truth tables'. The left-hand columns below indicate possible values of A, B, whereas the right-hand columns indicate the resulting value of the formula.

A	$\neg A$
F	T
T	F

A	B	$A \wedge B$	$A \vee B$
F	F	F	F
F	T	F	T
T	F	F	T
T	T	T	T

A	B	$A \to B$	$A \leftrightarrow B$
F	F	T	T
F	T	T	F
T	F	F	F
T	T	T	T

Every Boolean formula thus determines a *truth function* whose input is a collection of values (either T or F) for each proposition appearing in the formula, and whose output is either T or F. In fact, if we wish to be totally precise, we may consider the truth tables above as *definitions* of the Boolean operations.

Equivalent formulas Just as in English and Chinese, we may formulate the same sentence in multiple ways, there can be many different Boolean formulas which express the same statement. More precisely, we will say two Boolean formulas A and B are *logically equivalent* if they determine the same truth function. In this case, we will write

$$A \equiv B$$

In the analogy with algebraic equations, logical equivalence should be considered the analogue of the $=$ sign.

The most immediate example of a logical equivalence is something common in natural language too: the double negative. If I say I am "not not cold" it means I am cold. Likewise, for any proposition A, we have

$$\neg(\neg A) \equiv A.$$

As a slightly less trivial example, we may observe that we can write $A \to B$ by referring only to "simpler" symbols $\neg$ and $\vee$:

$$A \to B \equiv (\neg A) \vee B.$$

That is, A implying B is the same thing as saying that either A is false or B is true (or both).

The following equivalences are common and important enough that we record them here:

1. Symmetry:
   - $A \wedge B \equiv B \wedge A$
   - $A \vee B \equiv B \vee A$
2. Associativity:
   - $A \wedge (B \wedge C) \equiv (A \wedge B) \wedge C$
   - $A \vee (B \vee C) \equiv (A \vee B) \vee C$
3. Distributivity:
   - $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$
   - $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$
4. de Morgan's laws:
   - $\neg(A \wedge B) \equiv (\neg A) \vee (\neg B)$
   - $\neg(A \vee B) \equiv (\neg A) \wedge (\neg B)$
5. Contrapositive: $A \to B \equiv (\neg B) \to (\neg A)$

Exercise 2. *Check that the above Boolean formulas are logically equivalent by inspecting their truth tables.*

Warning 1. *In general we need to be careful to distinguish between two forms of equivalence:*

$$A \leftrightarrow B \tag{1.3}$$

and

$$\phi \equiv \psi \tag{1.4}$$

The expression (1.3) *is a Boolean formula stating that the propositions A and B are equivalent. The expression* (1.4)*, on the other hand, is not a Boolean formula (the symbol $\equiv$ is not part of the Boolean alphabet!). It represents the statement that the Boolean formulae ϕ and ψ are logically equivalent.*

1.5 Propositional logic: rules of inference

Deduction (also called inference) is the art of constructing new knowledge from existing knowledge.

Consider the Boolean formulas

$$A \to B \tag{1.5}$$

and

$$B \to C. \tag{1.6}$$

If we believe that both of these formulas are true, then we can *deduce* that

$$A \to C \tag{1.7}$$

is true.

Here is another example. Consider

$$\neg A \tag{1.8}$$

and

$$B \to A. \tag{1.9}$$

If we believe that both formulas are true, then we can deduce that B is false. We write these deductions as

$$\begin{array}{l} A \to B. \\ B \to C. \\ \hline \therefore A \to C. \end{array}$$

and

$$\begin{array}{l} \neg A. \\ B \to A. \\ \hline \therefore \neg B. \end{array}$$

So far, we are speaking intuitively. More formally, what we mean is that *every assignment of truth values to A, B, C which makes the first two formulas true will also make the third formula true.* We can think of this like a compound version of the truth tables of the above section.

Exercise 3. *Check the above examples using truth tables. In other words, check that if B is true, then at least one of the two formulas* (1.8) *or* (1.9) *must be false.*

This is an example of logical deduction. Although this kind of deduction is denoted by the special symbol $\vdash$ or $\models$ in formal logic, in practice we usually simply write (in English) something like

It follows from (1.8) and (1.9) that B is false.

Here is another example. If we believe

$$(A \vee B) \wedge \neg B \tag{1.10}$$

is true, then we can deduce that A is true.

Exercise 4. *Check this!*

Exercise 5. *Show that*

$$A \to B \tag{1.11}$$
$$B \to C \tag{1.12}$$
$$C \to A \tag{1.13}$$

is equivalent to

$$A \leftrightarrow B \leftrightarrow C. \tag{1.14}$$

There are a finite number of rules that allow us to deduce new Boolean formulas from old ones. We will not learn all of these rules, but you can find them listed on e.g. Wikipedia. You should not try to memorize them, and for practical purposes, your common sense will be enough.

1.6 Tautology and contradiction

There are two kinds of 'extreme' Boolean formulas: those which are always true, and those which are always false.

Example 1. *The formula*

$$A \leftrightarrow A$$

is always true, regardless of the truth-value of A. The same holds for

$$A \to A$$

and

$$A \vee \neg A.$$

We call such a formula a *tautology*.

Example 2. *The formula*

$$A \wedge \neg A$$

is always false, regardless of the truth-value of A.

We call such a formula a *contradiction*.

1.7 Why are we learning this?

Omar: say something about computer science?

In practice, you will not see many Boolean symbols used in your math classes, nor will you find many Boolean symbols in math papers. Mathematical reasoning is usually communicated using a language like English or Chinese (supplemented with appropriate notation), not using the Boolean language. Boolean is the language of computers: writing things in Boolean typically makes them harder for us humans to understand! However:

Any mathematician[2] *should be able to translate their reasoning from English/Chinese into Boolean.*

Just because you can, however, does not mean you should – just as paying for a new car using pennies would be incredibly impractical and irritate other people. Once you have some practice with Boolean logic, you can go back to writing your arguments in English or Chinese (or Irish, Ukrainian, or Urdu). You may find that you can express yourself more clearly and less ambiguously, having internalized the basic rules of Boolean logic.

In fact, Boolean logic is not powerful enough to capture all mathematical arguments. In the next section, we will learn about *first-order logic*, which is powerful enough for almost all of modern mathematics. Contrary to Boolean logic, the symbols of first-order logic will be frequently used in your classes and in math papers.

1.8 Practical examples of propositional logic

See the worksheet.

1.9 Predicate logic.

Consider the following argument:

All humans are mortal.
Confucius is human.

Therefore, Confucius is mortal.

This argument is difficult or impossible to state using only propositional logic.

Instead, we need the more powerful tools of *predicate logic*, which we introduce in this section. We will introduce the ingredients of predicate logic one by one:

- Variables $x, y, z, \ldots$.
- Predicates $P, Q, R, \ldots$.
- Quantifiers $\forall, \exists$.

[2] At least, English- or Chinese-speaking ones.

Variables Predicate logic deals with statements like 'All humans like S', or 'some humans dislike S'. In other words, we have a collection of things (humans in this example), and we are interested in whether some or all of them have some property. More generally, we want to make statements about things in some kind of collection of objects, without yet specifying which object we are talking about. We will use lower-case *variables* $x, y, z, ..$ to indicate the unspecified members of this collection.

Let's write $M(x)$ for 'x is mortal' and $H(x)$ for 'x is human'. Then we can write our original argument as

$$\begin{array}{l} \text{For all } x,\ H(x) \to M(x). \\ H(s). \\ \hline \therefore M(s). \end{array}$$

Predicates The word 'predicate' means 'property', and an important feature of predicate logic is that we can form a TRUE/FALSE proposition of the form

The object s has the property P.

For instance, let's write s for Confucius, and P for the property of being bearded. We write

$$P(s)$$

for the statement 'Confucius is bearded'. We can also apply M to other people: if we write p for Mencius, we can write

$$P(p)$$

for 'Mencius is bearded'.

Put another way: in propositional logic, the basic variables we considered were only capable of being either True (T) or False (F). In predicate logic, we can now consider another answer: it depends (on what x or some other variable is). We can think of predicates a little like "propositions depending on an arbitrary variable".

Quantifiers Why did we write 'for all x' in the first line? Why not just $H(x) \to M(x)$?

The formula $H(x) \to M(x)$ is neither true nor false: that depends on what x is. By adding 'for all x', we make it into a proposition which can be either true or false.

We use two special symbols for this, called *quantifiers*:

- $\forall$ (FOR ALL).
- $\exists$ (THERE EXISTS).

The formula

$$\forall x P(x)$$

means 'for all x, x has property P', whereas

$$\exists x Q(x)$$

means 'there exists x, such that x has property Q'. We can use these symbols to rewrite our original argument as

$$\begin{array}{l} \forall x(H(x) \to M(x)). \\ H(s). \\ \hline \therefore M(s). \end{array}$$

As with propositional logic, whenever we construct a sensible string of symbols using variables together with $\vee$, $\forall$, etc., we refer to it as a *predicate formula.* As before, we leave the grammar of this language to common sense.

Exercise 6. *Write the following as predicate formulas:*

1. *All couples have fights.*
2. *Everyone is reading at least two books.*
3. *My classmate has exactly one friend.*

The statement of "exactly one" existing is common enough in mathematics that we have a shorthand for it: $\exists!$. Using this symbol, we can also write the last statement as $\exists! x F(o, x)$.

Equivalence Just as we had a notion of logical equivalence in propositional logic, we also have a notion of logical equivalence between predicate formulas. However, it is slightly more complicated to define formally, so we will stick to our intuition when dealing with it: roughly speaking, two predicate formulas are logically equivalent if they hold for all possible substitutions.

Exercise 7. *Suppose P is a predicate. Rewrite*

$$\neg \forall x P(x)$$

using $\exists$.

Remark 2. *As a corollary, it is clear that $\exists x$ is the same thing as saying $\neg \forall x \neg$, and thus the symbols $\exists$ and $\forall$ are not independent.*

Consider the following argument:

My friend does not like cheese.
My friend is human.

Therefore, not all humans like cheese.

We write $C(x)$ for 'x likes cheese'. Then this becomes

$$\frac{\exists x(H(x) \wedge \neg C(x)).}{\therefore \neg \forall x(H(x) \to C(x)).}$$

Exercise 8. *Find a few examples of laws of predicate logic.*

Omar: what do you mean by laws

Of course, we can perfectly well allow our predicates to depend on multiple variables as well. For example, we could have the predicate $D(x, y)$ meaning "x dislikes y", or $L(x, y, z)$ meaning "x likes y with z on top".

Here are some important examples of equivalences in predicate logic:

1. Distributivity:
   - $\forall x(P(x) \wedge Q(x)) \equiv (\forall x P(x)) \wedge (\forall x Q(x))$
   - $\exists x(P(x) \vee Q(x)) \equiv (\exists x P(x)) \vee (\exists x Q(x))$
2. Quantifier negation laws:
   - $\neg \forall x P(x) \equiv \exists x \neg P(x)$
   - $\neg \exists x Q(x) \equiv \forall x \neg Q(x)$
3. de Morgan's laws:
   - $\exists x \neg(P(x) \vee Q(x)) \equiv \exists x(\neg P(x) \wedge \neg Q(x))$
   - $\forall x \neg(P(x) \wedge Q(x)) \equiv \forall x(\neg P(x) \vee \neg Q(x))$
4. Permutation:
   - $\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
   - $\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$

Exercise 9. *Translate the following sentences into English, and explain the difference between them.*

$$\forall x \exists y(x < y)$$

and

$$\exists y \forall x(x < y).$$

In particular, we see that permutation does ***not*** *hold for mixed quantifiers:*

$$\forall x \exists y P(x, y) \text{ and } \exists y \forall x P(x, y)$$

are not logicaly equivalent.

Exercise 10. *List some variants of the equivalences listed above which fail.*

Remark 3. *Note that predicate logic is not a separate system from propositional logic, but rather an enhancement of it. To recover propositional logic, roughly speaking we think of propositions as those predicates which have no dependence on variables, and ban all usage of the symbols* $\forall$ *and* $\exists$*.*

1.10 Practical examples of predicate logic

See the worksheet.

Chapter 2

Sets

Armed with predicate logic, we need one more ingredient to complete the language which we need to do almost all of mathematics. A good reference for this material is the book *Naive Set Theory* by Paul R. Halmos.

2.1 Naive set theory

Sets A *set* is a collection of things.

Example 3. *$A = \{$Peng Chau, Cheung Chau, Lantau$\}$, $B = \{1, 2, 3\}$ and $C = \{$Chinese, English$\}$ are sets.*

Mathematics is written in the language of sets. We write

$$x \in A$$

if x is contained in A. We call x an *element* of B. Thus, continuing the previous example, $1 \in B$. We write

$$x \notin B$$

if x is not in B. Thus $5 \notin B$.

We usually write elements in lowercase, and sets in uppercase. But we sometimes deviate from that convention.

Warning 2. *A* multiset *is a collection of things (just like a set), except that a multiset can contain multiple copies of a single thing, and it remembers how many copies it contains. For example, $A = \{1, 2, 3\}$ and $B = \{1, 1, 2, 3\}$ represent two different multisets. In this class, we will only be interested in sets, and almost never work with multisets.*

We write $A = B$ if the sets A and B have the same elements. We say A and B are *equal*. As far as set theory goes, there is no difference between A and B. We record this by the following grandiose-sounding 'axiom':

Axiom 1 (The Axiom of Extension)**.** *$A = B$ if and only if*

$$\forall x(x \in A \leftrightarrow x \in B).$$

Remark 4. *Wait a second - what* exactly *is a set? At this stage, you may complain: 'a collection of things' doesn't sound very precise! What kind of things? What kind of collections? What is a thing or a collection anyway? How can we do mathematics starting from such a vague concept?*

There are two answers to this question. The first is that, with a bit of practice, you will develop a very precise intuition for what is a set and what isn't.

If the first answer does not satisfy you, then congratulations! You are a formalist. To work formally with sets, we will write down nine 'axioms', such as the axiom of extension above. These axioms will tell us how to build and characterize the sets we need using expressions like $x \in X, y \notin X$. We need only these axioms, together with the language of predicate logic, to do all the set theory we will ever need.

Since this class is not an introduction to axiomatic logic, we will not cover all of the nine axioms. But we will see a few of them as we go.

Inclusions of sets: We write

$$A \subset B$$

if every element of A is contained in B, and call A a *subset* of B. We call A a *strict subset* of B if in addition, B contains an element not in A.

Exercise 11. *Prove that $A \subset B$ and $B \subset A$ implies $A = B$.*

Warning 3. *Some authors write $A \subseteq B$ instead of $A \subset B$, and write $A \subset B$ only when A is a strict subset.*

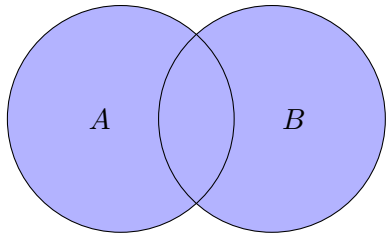
2.2 Making new sets

Even if we have our hands on some nice examples of sets, as greedy people we want to have more. To resolve this, we may construct new sets from old ones using various operations. We have:

1. $A \cup B$ (UNION):
   The *union* of two sets A and B is the set containing all elements that are in A or in B (or in both):

   $$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

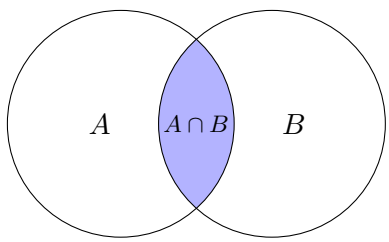
   Venn diagrams are often used to visually represent relationships between sets. For instance, the blue region below visualises the union of two sets.



2. $A \cap B$ (INTERSECTION):
   The *intersection* of two sets A and B is the set containing all elements that are in both A and B:

   $$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

   The Venn diagram visualizing the intersection looks like this.



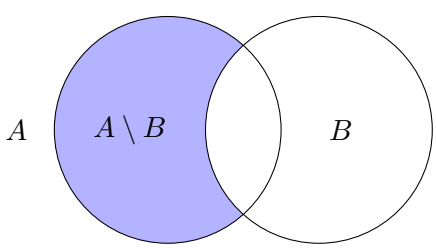
   Two sets are *disjoint* if $A \cap B = \emptyset$. What would the corresponding Venn diagram look like in this case?

3. $A \setminus B$ (COMPLEMENT and SET DIFFERENCE): The complement of a set $B \subset A$ is the set containing all elements of A that are not in B:

   $$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}.$$

   More generally, the *set difference* of two sets A, B, (not necessarily subsets)

   The Venn diagram visualizing the set difference looks like this.



4. $A \times B$ (PRODUCT):
   The *Cartesian product* of two sets A and B, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$:

   $$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

   For example, $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ is the Cartesian plane.

   Exercise 12. *Let the set A have exactly 5 elements, and the set B exactly 3 elements. How many elements does $A \times B$ have?*

   More generally, the product of n sets $A_1, A_2, \ldots, A_n$ is:

   $$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for each } i\}.$$

5. A^n (n-FOLD PRODUCT):
   As a special case of the above, the *n-fold Cartesian product* of a set A with itself n times is denoted A^n as a shorthand:

   $$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ times}} = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A \text{ for all } 1 \leq i \leq n\}.$$

   For instance, $\{0, 1\}^3 = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}$.

6. $P(A) = 2^A$. (POWER SET):
   The *power set* of a set A, denoted $\mathcal{P}(A)$ or 2^A, is the set of all subsets of A:

   $$\mathcal{P}(A) = \{X \mid X \subset A\}.$$

   If A has n elements, then $\mathcal{P}(A)$ has 2^n elements. For example,

   $$\mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}.$$

   The notation 2^A arises because there is a natural bijection between the power set and the set of all functions from A to the two-element set $\{0, 1\}$.

Disjoint union* $A \sqcup B$ (DISJOINT UNION):
The *disjoint union* of two sets A and B is given by $A \sqcup B = (A \times \{1\}) \cup (B \times \{2\})$. More generally, we can write the disjoint union of n sets A_n:

$$\bigsqcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} \{(x, i) \mid x \in A_i\}$$

The additional index is effectively "tagging" each element with its set of origin, so that we can remember which set it originally came from.

The Axiom of Comprehension Recall that in our discussion of predicate logic, we were somewhat vague about where variables could actually take values. For the purposes of this course, we will take them to represent elements of sets. With this in mind, we can state:

Axiom 2 (Axiom of Comprehension)**.** *Let P be a predicate and S a set. Then*

$$\{x \in S \mid P(x)\}$$

is a set.

2.3 The algebra of sets

Lemma 1. *Let A, B, C be sets. Then*

- $A \cup \emptyset = A$
- $A \cup B = B \cup A$ *(commutativity)*
- $A \cup (B \cup C) = (A \cup B) \cup C$ *(associativity)*
- $A \cup A = A$ *(idempotence)*
- $A \subset B$ *if and only if* $A \cup B = B$

Exercise 13. *State and prove the analogous properties for intersections.*

Lemma 2. *(Distributive laws) Let A, B, C be sets. Then*

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Exercise 14. *Prove or disprove the following statement:*
The equality $(A \cap B) \cup C = A \cap (B \cup C)$ holds true if and only if C is a subset of A.

Often, one is in a situation where everything is happening in some particular set (for example, our daily lives take place in $\mathbb{R}^3$). Let us therefore assume that all the sets below are subsets of a universal set E (here E stands for "everything"). Then for a set A, we denote by A^c the complement of A in E, i.e. $A^c = E \setminus A$ (note that the set E is dropped from the notation, so you should only use this when it is obvious).

Lemma 3. *Suppose $A \subset E$. Then:*

- $(A^c)^c = A$

- $\emptyset^c = E$, $E^c = \emptyset$
- $A \cap A^c = \emptyset$, $A \cup A^c = E$
- $A \subset B$ *if and only if* $B^c \subset A^c$

Lemma 4. *Suppose $A \subset E$ (de Morgan's laws)*

- $(A \cup B)^c = A^c \cap B^c$
- $(A \cap B)^c = A^c \cup B^c$

2.4 Functions

We will start with an informal discussion of functions, and then make it more precise using sets.

Informally, a *function*[1] f is a rule that takes an input[2] x and returns an output[3] $f(x)$.

Example 4. *You have learned many examples of functions in school, such as*

$$f(x) = x^2$$

and

$$g(x) = \sin(x).$$

The *domain* of f is the set of possible values of x. The *target* of f is the set of possible values of f. We write

$$f : X \to Y.$$

for a function with domain X and target Y.

The most important feature of a function is that it returns a *unique* output for a given input.

Warning 4. *Some people call such things* single-valued functions*, and also allow things called* multivalued *functions, which have multiple outputs. An example of such a multivalued function might be $\sqrt{x}$, with*

$$\sqrt{4} = \pm 2.$$

We will reserve the word function *for single valued functions, and we will not use the term 'multivalued function' in this class.*

[1] Synonyms: map, mapping
[2] Synonym: argument
[3] Synonym: value

The *graph* of f is the subset of $X \times Y$ defined by

$$\{(x, y) \mid y = f(x).\}$$

Knowing the graph is the same as knowing the function.

Question 3. *What kind of subsets of $X \times Y$ are graphs of functions?*

Here is one property satisfied by all graphs. Let f be a function. Then for each $x \in X$, there is a unique $y \in Y$ such that (x, y) lies in the graph of f. In fact, any subset $R \subset X \times Y$ satisfying this condition is the graph of a function: we can simply define $f(x)$ to be the unique y such that $(x, y) \in R$.

In the next subsection, we will turn this around and use graphs to give a rigorous *definition* of functions.

Set-theoretic definition of functions

Definition 1. *Let X, Y be sets. A function with domain X and target Y is a subset*

$$f \subset X \times Y$$

such that for all $x \in X$, there exists a unique $y \in Y$ such that $(x, y) \in f$. We write

$$y = f(x)$$

if $(x, y) \in f$.

We sometimes write $f : X \to Y$ and/or use the notation

$$x \to f(x).$$

Exercise 15. *Given sets A and B, show that there exists a set B^A consisting of all functions with domain A and target B.*

Examples of functions Every set comes with a seemingly trivial but very important function

$$\mathrm{id}_X : X \to X$$

which sends $x \mapsto x$, called the *identity* on X.

Exercise 16. *Give some other examples of functions.*

Image and preimage: Consider a function $f : X \to Y$. There are several closely related usages of the word *image*:

- *image of an element:* For $x \in X$, $f(x)$ is the image of x.
- *image of a subset:* For $A \subset X$,
$$f(A) = \{f(x) \mid x \in A\} \subset Y.$$
- *image of f:* The set $f(X)$ of all actual outputs:
$$f(X) = \{y \in Y \mid \exists x \in X \text{ with } f(x) = y\}$$
- *preimage:*[4] For $B \subset Y$,
$$f^{-1}(B) = \{x \in X \mid f(x) \in B\} \subset X.$$
Note: $f^{-1}(B)$ exists even if f is not invertible.
- For a single element $y \in Y$, we write $f^{-1}(y) = f^{-1}(\{y\})$.

Composition: Given $f : X \to Y$ and $g : Y \to Z$, the **composition** $g \circ f : X \to Z$ is defined by:
$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in X.$$
Composition is associative: $h \circ (g \circ f) = (h \circ g) \circ f$ when defined.

Guessing the domain of a function In practice, a function is often given without specifying its domain explicitly. For instance, consider the function $\frac{1}{x}$. This function takes as input a number, and outputs a number. Depending on context, this might be a rational number or a real number. However, in all of these cases, it does *not* take the argument $x = 0$; we say it is not *defined* at 0.

So we might interpret $f = \frac{1}{x}$ as a function with domain $\mathbb{R} \setminus 0$ or $\mathbb{Q} \setminus 0$, depending on context.

Injectivity, surjectivity, bijectivity: $f : X \to Y$ is *injective* or *one-to-one* if distinct inputs yield distinct outputs:
$$\forall x_1, x_2 \in X,\ f(x_1) = f(x_2) \implies x_1 = x_2.$$
Equivalently, $\forall y \in Y$, $f^{-1}(y)$ contains at most one element.

$f : X \to Y$ is *surjective* or *onto* if every element of Y is an output:
$$\forall y \in Y,\ \exists x \in X \text{ such that } f(x) = y.$$
Equivalently, $f(X) = Y$. Finally, $f : X \to Y$ is *bijective* if it is both injective and surjective.

Exercise 17. *Give some examples and non-examples of injective and surjective functions.*

[4] Synonyms: inverse image, fibre

Inverses: For $f : X \to Y$, bijective we can define the notion of an inverse function. We say an *inverse function* $g : Y \to X$ exists if and only if there is a function g, denoted f^{-1}, such that

- $f^{-1} \circ f = \mathrm{id}_X$ (left inverse).
- $f \circ f^{-1} = \mathrm{id}_Y$ (right inverse).

Exercise 18. *Show that the inverse function of f, if it exists, is unique.*

Exercise 19. *Show that an inverse function of f exists if and only if f is a bijection.*

Remark 5. *Beware of a slight notational overlap. The symbol f^{-1} has two meanings:*

- *For any function $f : X \to Y$, $f^{-1}(B)$ denotes the preimage of $B \subset Y$.*
- *For a bijective function f, f^{-1} also denotes the inverse function.*

The interpretation should essentially always be clear from the context.

2.5 Cardinality

Cardinality is a fancy way of saying 'size'.

Definition 2. *Two sets X and Y* have the same cardinality *if there is exists a bijection*

$$f : X \to Y.$$

We write

$$|X| = |Y|.$$

We will see later that this concept is more subtle than it appears.

2.6 Operations

Definition 3. *Let S be a set. A function $f : S^n \to S$ is called an n-ary operation.*

Depending on the context, there are different standard notations for such things:

Example 5. *Addition of numbers is a 2-ary operation, which we write as*

$$(a, b) \to a + b.$$

Example 6. *Exponentiation of numbers is a 1-ary operation, which we write as*

$$x \to e^x.$$

Example 7. *We can even talk about 0-ary operations: these are functions to S with no arguments at all! That is, a 0-ary operation*

$$f : \emptyset \to S$$

can be thought of as just an element of S.

2.7 Relations

A *relation* on a pair of sets X, Y is a subset

$$R \subset X \times Y.$$

We write

$$xRy$$

to mean

$$(x, y) \in R.$$

Example 8. *Let X be the set of countries and Y be the set of cities in the world. Then*

$$\{(x, y) \in X \times Y \mid y \text{ is the capital city of } x\}$$

is a relation. Whether it is a function depends on which era we are in, as many states had summer capitals historically; thanks to air conditioning, it is most likely a function today.

If $X = Y$, we simply refer to it as a relation on X (or Y) . An example of a relation on $\mathbb{Z}$ is

Omar: is this what you want

$$\left\{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid y^2 = x.\right\}$$

The *domain* of R is the set of $x \in X$ such that there exists $y \in Y$ with xRy. The *image* of R is the set of $y \in Y$ such that there exists $x \in X$ with xRy.

Exercise 20. *Find the domain and range of the above relation.*

Clearly, a function is a special case of a relation, but with the restriction that a given element in X can only be related to one element of Y.

Example 9. *Show that the relation on $\mathbb{R}$*

$$\left\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x^2 + y^2 = 1\right\}$$

does not define a function.

2.8 Relations from a set to itself

Relations $R \subset X \times X$ from a set to itself play a special role. Here are some extra properties they can satisfy.

Definition 4. *We say a relation $R \subset X \times X$ is*

- transitive *if*

$$\forall x, y, z : xRy \wedge yRz \to xRz.$$

- reflexive *if*
$$\forall x : xRx.$$
- symmetric *if*
$$\forall x, y : xRy \leftrightarrow yRx$$
- antisymmetric *if*
$$\forall x, y : xRy \wedge yRx \rightarrow x = y.$$

Exercise 21. *Come up with a few examples of relations which satisfy and fail each property.*

2.9 Partial orders

A very important example of a relation is the following.

Definition 5. *A relation $R \subset X \times X$ is a partial order*[5] *if it is reflexive, antisymmetric, and transitive.*

In this case, we usually write $x \preceq y$ instead of xRy, and $x \prec y$ if $x \preceq y$ and $x \neq y$.

Example 10. *Let $X = \{a, b, c\}$ and $P = \mathcal{P}(X)$. Define:*

$$A \preceq B \quad \textit{if and only if} \quad A \subseteq B$$

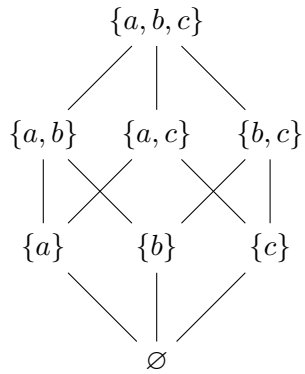
It is easy to see that the axioms are satisfied.

When feasible, we can depict a partial order using *Hasse diagram*. This is a graphical representation of a partial order where:

1. Elements are drawn as points (nodes)
2. If $a \prec b$ (i.e., $a \preceq b$ and $a \neq b$), then b appears **above** a
3. A line is drawn from a to b if $a \prec b$ and there is **no** c such that $a \prec c \prec b$ (i.e., b *covers* a)
4. All other relations are implied by transitivity (so we don't draw them)

We may draw the Hasse diagram of the above example as:

[5] Also partially ordered set, or poset for short.



Exercise 22. *Let $P = \{1,2,3\}$ and define $a \preceq b$ if $a \leq b$ (usual order)* ***except*** *also define $1 \preceq 3$ and $3 \preceq 1$. Show that this is not a partial order: which property does it fail to satisfy?*

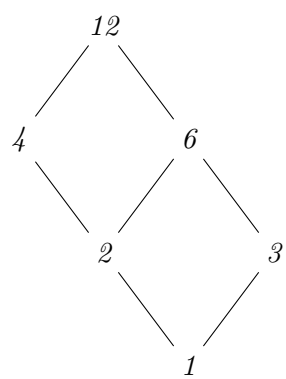
Example 11. *Let $X = \mathbb{Z}_+$ (positive integers) and define:*

$$a \preceq b \quad \textit{if and only if} \quad a \mid b \ (a \textit{ divides } b)$$

We check that this is a partial order:

- Reflexive: $a \mid a$ *since* $a = a \cdot 1$
- Antisymmetric: *If $a \mid b$ and $b \mid a$, then $a = b$ (for integers, up to sign, but here all positive)*
- Transitive: *If $a \mid b$ and $b \mid c$, then $a \mid c$ (try writing this out in predicate logic)*

Example 12. *Let $X = \{1,2,3,4,6,12\}$ and with the divisibility partial order as above. The Hasse diagram can be drawn as:*



2.10 Equivalence relations

The other fundamental example of a relation is that of an equivalence relation.

Definition 6. *A relation $R \subset X \times X$ is an* equivalence relation *if it is reflexive, symmetric, and transitive.*

Example 13. *Let $X = \{$people in this class$\}$ and define xRy if x and y live in the same building. Then R is an equivalence relation.*

Exercise 23. *Give an example of something that fails one of each property but satisfies the other two.*

Definition 7. *Let $m \in \mathbb{Z}$, $m > 0$. For integers $a, b \in \mathbb{Z}$, we say a and b are* congruent modulo m*:*

$$a \equiv b \pmod{m}$$

if $m \mid (a - b)$ (i.e., $a - b$ is divisible by m).

Equivalently: $a \equiv b \pmod{m} \iff a = b + km$ for some $k \in \mathbb{Z}$.

Exercise 24. *Show that congruence modulo m is an equivalence relation.*

Example 14.

$$a \equiv 0 \pmod{2} \leftrightarrow a \text{ is even}$$
$$a \equiv 1 \pmod{2} \leftrightarrow a \text{ is odd}$$

Definition 8. *Let $R \subset X \times X$. An* equivalence class *is a nonempty subset $S \subset X$ such that*

$$x \in S \wedge xRy \to y \in S$$

and

$$x, y \in S \to xRy.$$

In other words, an equivalence class is a subset of elements of X, all of whom are related to eachother, and with no "missing" related elements.

Definition 9. *Let R be an equivalence relation on X, and let $x \in X$. We define the* equivalence class *of x to be the subset $[x]_R \subset X$ given by*

$$\{y \mid xRy\}.$$

By reflexivity, it is immediate that for any $x \in X$, $x \in [x]_R$.

Proposition 1. *A set with an equivalence relation is the union of its equivalence classes, all of which are pairwise disjoint.*

Proof. Let $x \in X$. Then $x \in [x]_R$. This shows that X is a union of equivalence classes.

We now show that any two equivalence classes are equal or disjoint. Suppose $S, T \subset X$ are equivalence classes. Let $x \in S \cap T$. Let $y \in S$. Then xRy (because S is an equivalence class) and therefore $y \in T$ (because T is an equivalence class). This shows $S \subset T$. By the same argument, $T \subset S$ and thus $T = S$. $\square$

We sometimes say that the equivalence classes *partition* the set.

As a matter of notation, an equivalence relation R is often denoted instead by $\sim$, so that xRy is written $x \sim y$. The next definition will be very useful later.

Definition 10. *Let X be a set, and $\sim$ an equivalence relation on it. The set of equivalence classes is called the* quotient set*, denoted by $X/\sim$.*

Mykola: added some facts collected from Halmos

Chapter 3

The integers

3.1 A mathematician's view of the integers

Consider the following sentence:

Any odd integer greater than 5 *may be expressed as the sum of three primes.*

We want to think about such statements using the language of logic and set theory. Our first task is to understand the set of all natural numbers. We make the following informal definition:

Definition 11. *Let*

$$\mathbb{N} = \{0, 1, 2, 3, 4, 5, ...\} \tag{3.1}$$

be the set of all natural numbers.

This definition is *informal* because we have not said exactly what the elements of this set are. Instead, we use the notation ... to *suggest* that aside from $0, 1, 2$, we should also include other elements (such as 3 and 4), and we think the reader knows how to guess these elements.

Another seemingly frivolous objection: different people may use different symbols for numbers:

Definition 12. *Let*

$$N = \{0, I, II, III, IV, V, ...\}$$

be the set of all Roman numbers.

Of course, this is not a practical problem : we can easily translate between the two. A more serious issue: how would you explain to a computer, or an alien visiting from the rings of Saturn, what those mysterious (...) are hiding?

In the end, the main problem with our informal definition is that it does not reveal what makes the natural numbers the natural numbers. It does not capture their essence!

3.2 A formal definition: Peano's axioms

Let's try again. Suppose $\mathcal{N}$ is a set together with:

- An element $0 \in \mathcal{N}$.
- A function $S : \mathcal{N} \to \mathcal{N}$.

Definition 13. *We say the triple of data $(\mathcal{N}, 0, S)$ (or just $\mathcal{N}$ for short) is a* model of the natural numbers *if:*

1. *If $S(x) = S(y)$ for $x, y \in \mathcal{N}$, then $x = y$ (S is injective).*
2. *For all $x \in \mathcal{N}$, $S(x) \neq x$.*
3. *There is no $x \in \mathcal{N}$ such that $S(x) = 0$.*
4. *(Principle of Induction) Suppose a predicate[1] P is true of 0, and that whenever P is true of x, it is also true of $S(x)$. Then P is true for all $x \in \mathcal{N}$.*

Exercise 25. *Show that the set $\{0, 1, 2\}$ with $S(0) = 1$, $S(1) = 2$, $S(2) = 0$* does not *satisfy all Peano axioms. Which axioms fail?*

We are used to thinking of these as properties or structures of the natural numbers. Now we will reverse our perspective and think of them as a *definition* of the natural numbers.

Compared with our informal definition, this has a number of drawbacks. It is hard to understand at first, and uses unfamiliar notation. It seems overly complicated for the apparently simple task of defining the natural numbers.

It has the benefit, however, of being completely defined in terms of sets and predicate logic. No magical intuition or fundamentally new concepts are required, and it becomes a construction reducible to more primitive ingredients.

3.3 Defining arithmetic operations: $+$ and $\cdot$

We would like to define the usual operations we are familiar with on integers. First, we define $+$:

$$\begin{cases} a + 0 = a \\ a + S(b) = S(a + b) \end{cases}$$

and likewise with the order reversed.

This is a recursive definition: we begin with a "base case" of adding 0 to itself or to other numbers, and to add something to a we first express it as a successor, then reduce the problem to computing the successor of a "smaller" number, $a + b$. Repeatedly applying this definition, we eventually get back down to 0.

[1] Strictly: a unary predicate, that is, with one argument.

Exercise 26. *Show this formally using Peano's axioms.*

It is instructive to compute an example to see exactly how this works. Let's try to catch up with elementary school students, and show $2 + 3 = 5$.

Example 15. *Let* $2 = S(S(0))$, $3 = S(S(S(0)))$. *Then:*

$$\begin{aligned} 2 + 3 &= S(S(0)) + S(S(S(0))) \\ &= S(S(S(0)) + S(S(0))) \\ &= S(S(S(S(0)) + S(0))) \\ &= S(S(S(S(S(0)) + 0))) \\ &= S(S(S(S(S(0))))), \end{aligned}$$

a number commonly denoted as 5.

Exercise 27. *Show that for all* $n \in \mathbb{N}$, $0 + n = n$.

As with addition, we define $\cdot$ recursively too:

$$\begin{cases} a \cdot 0 = 0 \\ a \cdot S(b) = a + (a \cdot b) \end{cases}$$

This is well-defined again by the principle of induction.

Exercise 28. *Calculate* $2 \cdot 3$.

Exercise 29. *Show from the definitions that addition and multiplication are commutative and associative.*

Exercise 30. *Show that for any* $a, b, c \in \mathbb{N}$, *we have the distributivity property:*

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

3.4 Principle of Induction

Having played around with some of the consequences, it is hopefully now clear what we have done: we have taken the idea of mathematical induction and turned it into a definition. In essence, this is the only property that one ever needs or uses about the natural numbers. It is less important to us what they actually *are*, and more important how they are *related to one another*, in order that we may *prove properties*. This is more generally a common principle of (much of) modern mathematics, and you should contemplate it.

Exercise 31. *Let $M = \{0, 1, 2, \ldots\} \cup \{a, b, c\}$, with the usual successor function*

$$S(0) = 1, S(1) = 2, \ldots$$

for the numerical elements, and define

$$S(a) = b, \quad S(b) = c \quad S(c) = a.$$

Show that $(M, 0, S)$ satisfies all of the conditions to be a model of the natural numbers except *the principle of induction.*

3.5 From $\mathbb{N}$ to $\mathbb{Z}$

At some point, one encounters equations in which only natural numbers appear, like $1 = x + 2$, but which have no solution in the natural numbers. We would like to extend our number system to allow for these more exotic "numbers" which, as the naming suggests, are not quite as natural (I have certainly never seen -1 apples)[2].

To do this, we will first take a set that is too big, and put an equivalence relation on it. The set of equivalence classes will then be the thing we are looking for. That sounds rather elaborate, but the motivation for doing this is the following: we are trying to represent "the solution to $1 = x + 2$". However, whatever this thing is, it is also the solution to $2 = x + 3$, and the solution to $3 = x + 4$. We can therefore try to think of this entire collection of equations, indexed by $(1, 2), (2, 3), (3, 4), \ldots$ as representing the same thing, and *define* that to be -1.

Definition as a quotient set More precisely, and formally, we consider an equivalence relation $\sim_{\mathbb{Z}}$ on the set $X = \mathbb{N}^2$ given by

$$(a, b) \sim_{\mathbb{Z}} (c, d) \text{ if } a + d = b + c.$$

Exercise 32. *Check that this is an equivalence relation.*

We define $\mathbb{Z} := X/\sim_{\mathbb{Z}}$, the set of equivalence classes:

$$[(a, b)] := \{(c, d) \in X \mid (c, d) \sim_{\mathbb{Z}} (a, b)\}.$$

Thus, an *integer* is an equivalence class of the relation $\sim_{\mathbb{Z}}$ on $X = \mathbb{N}^2$.

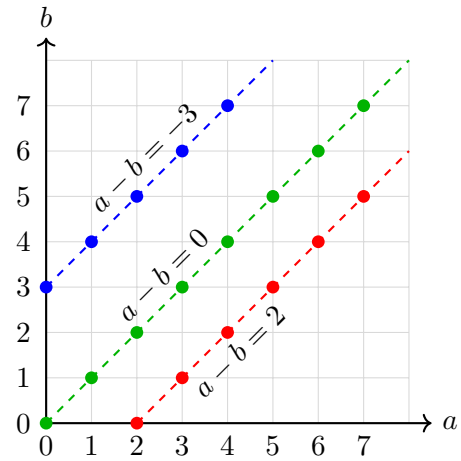
This looks a little bit funny. Aren't the integers supposed to contain the naturals? Where are they?

In fact, we can embed the familiar $\mathbb{N}$ into $\mathbb{Z}$ via the map:

$$n \mapsto [(n, 0)].$$

Check: If $m \neq n$, then $m + 0 \neq n + 0$, so $[(m, 0)] \neq [(n, 0)]$. Thus the map is injective, so we have no problem identifying $n \in \mathbb{N}$ with $[(n, 0)] \in \mathbb{Z}$.

[2] According to Leopold Kronecker, "Die ganzen Zahlen hat der liebe Gott gemacht, alles andere ist Menschenwerk" ("God made the integers, all else is the work of man"). Actually, here it is better to say that the integers are also the work of man, being formed out of the naturals.



Since we're thinking of the equivalence class of (a, b) to mean $a - b$, we can introduce standard terminology: *positive* if $a > b$, *negative* if $a < b$, and *zero* if $a = b$. Let us practice a bit with our definition to show a simple result, namely that this actually makes sense:

Proposition 2. *Every element of $\mathbb{Z}$ is well-defined to be either positive, negative, or zero.*

Proof. We need to show that the properties are well-defined on equivalence classes. Suppose $a + d = b + c$ and $a > b$, meaning there exists $k \in \mathbb{N}_+$ such that $a = b + k$. Then

$$b + k + d = b + c$$

so that $c = d + k$ with $k \in \mathbb{N}_+$, meaning (c, d) is also positive. The same argument works for the other cases. $\square$

Exercise 33. *Define subtraction on $\mathbb{Z}$.*

Arithmetic operations We had a notion of addition and multiplication for natural numbers, and we certainly expect we should have one for integers too. But we need to define it in terms of our equivalence classes now. For integers $[(a, b)]$ and $[(c, d)]$, define:

$$[(a, b)] + [(c, d)] := [(a + c,\ b + d)].$$

We need to check this is well-defined. If $(a, b) \sim_{\mathbb{Z}} (a', b')$ and $(c, d) \sim_{\mathbb{Z}} (c', d')$, then

$$(a + c,\ b + d) \sim_{\mathbb{Z}} (a' + c',\ b' + d').$$

Proof. Since $a + b' = b + a'$ and $c + d' = d + c'$, add them:

$$(a + b') + (c + d') = (b + a') + (d + c').$$

Rearrange:

$$(a+c)+(b'+d') = (b+d)+(a'+c').$$

This exactly means $(a+c,\ b+d) \sim_{\mathbb{Z}} (a'+c',\ b'+d')$. $\square$

Similarly, we define multiplication by

$$[(a,b)] \cdot [(c,d)] := [(ac+bd,\ ad+bc)].$$

Exercise 34. *Show this is well-defined.*

As we might expect from our intuition, the usual properties we showed hold for naturals also hold for integers:

Exercise 35. *Show that addition and multiplication of integers are commutative, associative, and satisfy distributivity.*

3.6 From $\mathbb{Z}$ to $\mathbb{Q}$

As we learned many years ago, there are situations in which even integers do not suffice to do mathematics, ranging from the mundane (splitting an apple), to the purely mathematical (we want to be able to solve equations like $1 = 2x$). Which one you find more motivating is a matter of taste, but the necessity of adding more numbers to our system becomes clear.

Just as we defined the integers out of the naturals, we will define rational numbers using a certain equivalence class on pairs of integers (now you may be starting to get a sense of what Kronecker meant). Let

$$Y = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) = \{(a,b) \mid a, b \in \mathbb{Z}, b \neq 0\}.$$

Define a relation $\sim_{\mathbb{Q}}$ on Y by:

$$(a,b) \sim_{\mathbb{Q}} (c,d) \iff a \cdot d = b \cdot c$$

where the $\cdot$ refers to multiplication as defined for $\mathbb{Z}$. Notice that this is none other than formalizing what we know about fractions as children!

Exercise 36. *Show that $\sim_{\mathbb{Q}}$ is an equivalence relation on Y.*

Definition 14 (Rational Numbers)**.** *The set of* rational numbers *is:*

$$\mathbb{Q} = Y/\sim_{\mathbb{Q}} = \{[(a,b)] \mid (a,b) \in Y\}$$

where $[(a,b)] = \{(c,d) \in Y \mid (a,b) \sim_{\mathbb{Q}} (c,d)\}$ is the equivalence class of (a,b).

We write $[(a,b)]$ as $\frac{a}{b}$, and use the standard notations like $0 = [(0,1)]$, $1 = [(1,1)]$, etc.

Once again, we want to be able to add and multiply rational numbers, but it is not immediately obvious how to do this. The solution, as before, is to simply codify what we already know about rational numbers in our hearts.

For addition, we define

$$[(a,b)] + [(c,d)] := [(ad+bc,\ bd)].$$

If $[(a,b)]$ represents $\frac{a}{b}$ and $[(c,d)]$ represents $\frac{c}{d}$, then:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}.$$

Thus $[(ad+bc,\ bd)]$ agrees with our intuition. We check it is well-defined:

Proof. Assume $(a,b) \sim_{\mathbb{Q}} (a',b')$ (so $ab' = ba'$) and $(c,d) \sim_{\mathbb{Q}} (c',d')$ (so $cd' = dc'$). We need to show:

$$(ad+bc,\ bd) \sim_{\mathbb{Q}} (a'd'+b'c',\ b'd').$$

That is:

$$(ad+bc)(b'd') = (bd)(a'd'+b'c').$$

From $ab' = ba'$ and $cd' = dc'$, compute:

$$\begin{aligned}(ad+bc)(b'd') &= adb'd' + bcb'd' \\ &= (ab')(dd') + (cd')(bb') \\ &= (ba')(dd') + (dc')(bb') \\ &= bd(a'd'+b'c').\end{aligned}$$

□

Likewise, we define multiplication:

$$[(a,b)] \cdot [(c,d)] := [(ac,\ bd)].$$

motivated by our usual intuition for multiplication of rationals.

Exercise 37. *Show that this is well-defined.*

Exercise 38. *Show that addition and multiplication of rationals are commutative, associative, and satisfy distributivity.*

The rational numbers are great, and they have a very nice feature which we will later refer to as forming a *field*: every nonzero element can be "inverted" (i.e. multiplied by something to get 1). This is extremely important for doing things like number theory. Let's prove it:

Proposition 3. *Given nonzero rational number r, there exists a unique rational number r^{-1} such that $r \cdot r^{-1} = 1 = [(1,1)]$.*

Proof. Let $q = [(a, b)] \neq 0$, so $a \neq 0$. Define $q' = [(b, a)]$ (valid since $a \neq 0$). Then

$$q \cdot q' = [(a, b)] \cdot [(b, a)] = [(ab, ba)].$$

Since integers commute, $ab = ba$, so $(ab, ba) \sim_{\mathbb{Q}} (1, 1)$ because $(ab) \cdot 1 = (ba) \cdot 1$. Hence

$$[(ab, ba)] = [(1, 1)] = 1.$$

Thus q' is a multiplicative inverse for q.

We need to check that it's unique. Assume $[(b, a)] \cdot [(a, b)] = 1$ and $[(b', a')] \cdot [(a, b)] = [(1, 1)]$, i.e., $ab' = a'b$. This is exactly the condition for $(b', a') \sim_{\mathbb{Q}} (b, a)$. Hence the inverse map is unique. □

Irrationality of $\sqrt{2}$ Nice as the rationals are, they are not enough for all purposes. In fact, thousands of years ago mathematicians were shocked to discover that[3] despite the equation containing nothing but natural numbers (not even having used integers or rationals!):

Theorem 1. *There is no rational solution to $x^2 = 2$.*

Before proving it, let us show:

Lemma 5. *If p is an integer and p^2 is even, then p is even.*

Proof. We prove the contrapositive: If p is odd, then p^2 is odd.
Let $p = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$p^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is odd. Hence, if p^2 is even, p cannot be odd; so p is even. □

We also need to introduce terminology you may be familiar with: for two integers $p, q \in \mathbb{Z}$, their *greatest common divisor* $\gcd(p, q)$ is the largest (positive) integer d such that $d|p$ and $d|q$. Now we proceed with the main result.

Proof of Theorem 1. Suppose there exists a rational number x such that $x^2 = 2$. Since x is rational, we can write

$$x = \frac{p}{q},$$

where $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$ (i.e., the fraction is in lowest terms – this can always be done by dividing top and bottom a finite number of times[4]).

[3] In fact, some of the earliest known mathematics done by humans concerned this mysterious number "$\sqrt{2}$". Some of its digits were computed millenia ago in the Middle East, India, Africa, Europe, China, and probably elsewhere, before anything was known of irrational numbers.

[4] Proof: if $\gcd(p, q) = d > 1$, then $(p, q) \sim_{\mathbb{Q}} (\frac{p}{d}, \frac{q}{d})$ with $\frac{p}{d}, \frac{q}{d}$ both having fewer divisors (counted with multiplicity) than p, q. After doing this a finite number of times, there are no common divisors left except 1.

From $x^2 = 2$ we have

$$\left(\frac{p}{q}\right)^2 = 2 \quad \Rightarrow \quad \frac{p^2}{q^2} = 2 \quad \Rightarrow \quad p^2 = 2q^2. \tag{1}$$

Equation (1) tells us that $p^2 = 2q^2$, so p^2 is even. From the lemma, since p^2 is even, p is even. Thus we can write $p = 2k$ for some $k \in \mathbb{Z}$.

Substitute $p = 2k$ into equation (1):

$$(2k)^2 = 2q^2 \quad \Rightarrow \quad 4k^2 = 2q^2 \quad \Rightarrow \quad q^2 = 2k^2. \tag{2}$$

Equation (2) says $q^2 = 2k^2$, so q^2 is even. Applying the lemma again, q is even.

We have shown both p and q are even. This means $2 \mid p$ and $2 \mid q$, so $\gcd(p, q) \geq 2$. But this contradicts our initial choice that $\gcd(p, q) = 1$. $\square$

The purpose of the next chapter will be to remedy problems like this.

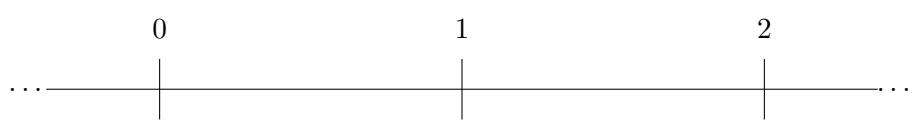
Chapter 4

The reals

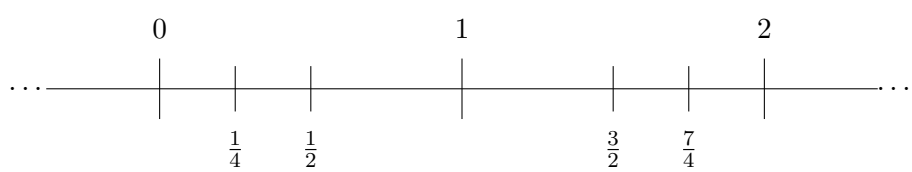
Now when I was a little chap I had a passion for maps. I would look for hours at South America, or Africa, or Australia, and lose myself in all the glories of exploration. At that time there were many blank spaces on the earth, and when I saw one that looked particularly inviting on a map (but they all look that) I would put my finger on it and say, 'When I grow up I will go there.'

Marlow, Heart of Darkness

Imagine an infinite ruler with integer spacings:



The rational numbers live between the integers:



As we add more and more rational numbers, our ruler gets more and more crowded.

No matter how many rationals we add, however, we can still find some blank spots. Our goal is to understand these mysterious blank spots: we call them the *real numbers*.

Unlike the integers or the rational numbers, calculations with real numbers are not daily features of human life. You may not have a clear idea of what real numbers really are. You might have seen heard:

- A real number is a point on the real line.
- A real number is something that looks like $\sqrt{2}$ or $\pi/3$.
- A real number is something that looks like

  $$10.234890234809234....$$

  In other words, it is given by a (potentially infinite) decimal expansion.

The first two answers are pretty vague. You may feel that the third answer is perfectly good: you already know how to add and multiply (potentially infinite) decimal expansions. However, there are a few problems with trying to define real numbers this way.

- A given real number may have many different decimal expansions. You may have learned in high school that

  $$0.999999.... = 1.$$

  As we shall see, one can fix this problem by considering decimal expansions up to a certain equivalence relation. But this redundancy suggests there might be a better, more direct definition.

- This strategy forces us to make the artificial choice of working in base 10, rather than for instance base 2 or base 3.

  We want a construction that does not force us to make this arbitrary choice.

- Perhaps the most serious problem with this strategy is that it does not clarify what is so useful and special about the real numbers.

In this chapter, we will answer the third point by writing down the basic properties which characterize the reals, and give a beautiful construction of them due to Dedekind.

The key to unlocking the reals is a structure we have not made much use of so far: the standard order $\leq$ on the rationals. We begin by studying it in more detail in the next section.

4.1 Total orders

Definition 15. *A set X with a partial order $\leq$ is* totally ordered *or* linearly ordered *if for all $x, y \in X$ either $x \leq y$ or $y \leq x$.*

For example, consider the usual $\leq$ relation on $\{1, 2, 3, 4\}$.

$$\begin{tikzcd}
4 \arrow[d, dash] \\
3 \arrow[d, dash] \\
2 \arrow[d, dash] \\
1
\end{tikzcd}$$

This is a total order, since any two elements can be compared.

For a non-example, consider $\{1, 2, 3, 6\}$ with the partial ordered defined by $x \leq y$ if x divides y. We can draw the Hasse diagram as follows:

$$\begin{tikzcd}
 & 6 \arrow[dl, dash] \arrow[dr, dash] & \\
2 \arrow[dr, dash] & & 3 \arrow[dl, dash] \\
 & 1 &
\end{tikzcd}$$

This is not a linear order, since 2 and 3 cannot be compared.

4.2 The standard order on $\mathbb{Z}$ and $\mathbb{Q}$

Previously, we defined subtraction for integers, and the notion of positivity and negativity. A similar construction can be made for rationals: we say $\frac{a}{b}$ is *positive* if the integers $a, b \in \mathbb{Z}$ are both positive or both negative, *negative* if one is positive and the other is negative, and *zero* if $a = 0$. We then define the standard order you are familiar with

$$x \leq y \text{ if } 0 \leq y - x$$

where we declare $0 \leq a$ if a is positive or zero. We can take $a \in \mathbb{Z}$ or $a \in \mathbb{Q}$, so that the standard order gives a partial order on $\mathbb{Z}$ or $\mathbb{Q}$.

Proposition 4. *The standard order $\leq$ on $\mathbb{Z}$ or $\mathbb{Q}$ is a total order.*

Proof. We have $x \leq y$ if and only if $0 \leq y - x$, and $x \geq y$ if and only if $0 \geq y - x$. Hence the claim follows for $\mathbb{Z}$ from the fact that every integer is either positive, negative or zero. The argument for rationals is similar. $\square$

Definition 16. *Let $(X, \leq_X)$ be a partially ordered set, and let $Y \subset X$ be a subset. We define the* restricted *partial order $\leq_Y$ on Y by*

$$y \leq_Y y' \leftrightarrow y \leq_X y'.$$

Usually, we drop the subscripts and simply write $\leq$ for both orders.

Exercise 39. *Suppose $(X, \leq)$ is totally ordered. Then any $Y \subset X$ equipped with the restricted order is also totally ordered.*

This gives us many examples of totally ordered sets, given by arbitrary subsets of the integers or rational numbers.

Of course, the rationals are not just an ordered set: they also have two binary operations $(+, \cdot)$. These play nicely with the order, as follows:

Proposition 5. *Let $x, y, z \in \mathbb{Q}$.*

- *If $x \leq y$, then $x + z \leq y + z$.*
- *If $x \geq 0$ and $y \geq 0$, then $x \cdot y \geq 0$.*

Exercise 40. *Prove it!*

4.3 Some fancy words

Let's take stock of what we know about $\mathbb{Z}$ and $\mathbb{Q}$, in order to see what is missing to define $\mathbb{R}$. This section is purely motivational.

Both $\mathbb{Z}$ and $\mathbb{Q}$ are sets equipped with distinguished elements $0, 1 \in \mathbb{Q}$ and binary operations $+, \cdot : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$. The pair of operations $(+, \cdot)$ are associative, commutative and distributive, and satisfy

$$0 + x = x \text{ and } 1 \cdot x = x \text{ for all } x.$$

Both $\mathbb{Z}$ and $\mathbb{Q}$ have additive inverses:

$$\text{For all } x, \text{ there exists } y \text{ such that } x + y = 0.$$

The rationals satisfy an additional property:

For all $x \neq 0$, there exists y such that $x \cdot y = 1$.

In other words, every element in $\mathbb{Q}$ has a multiplicative inverse.

Definition 17. *We call a set F equipped with elements $0, 1 \in F$ and binary operations $+, \cdot$ satisfying all the above properties a* field.

Hence, we can summarize almost everything we know so far about $\mathbb{Q}$ by saying '$\mathbb{Q}$ is a field'. Note, however, that the definition of field makes no mention of the order $\leq$ on $\mathbb{Q}$. Let's fix that.

Definition 18. *An* ordered field *is a field equipped with a total order $\leq$ satisfying:*

- *If $x \leq y$, then $x + z \leq y + z$.*
- *If $x \geq 0$ and $y \geq 0$, then $x \cdot y \geq 0$.*

By Proposition 5, $\mathbb{Q}$ is an ordered field. The reals will also be an ordered field. They will satisfy one extra property with respect to this order, which $\mathbb{Q}$ does not: *completeness*. The next few sections will build up to a definition of this property.

4.4 Upper bound, lower bound, infimum and supremum

In mathematics, interesting things tend to happen at the extremes - the biggest, the smallest, the first, the last. We need some words to describe these extremes.

Let Y be a partially ordered set.

- An element $y \in Y$ is a *maximum* for Y if for all $y' \in Y$, $y' \leq y$.
- An element $y \in Y$ is a *minimum* for Y if for all $y' \in Y$, $y' \geq y$.

Let X be a partially ordered set, and let $Y \subset X$ be a subset.

- An element $x \in X$ is an *upper bound* for Y if for all $y \in Y$, $y \leq x$.
- An element $x \in X$ is a *lower bound* for Y if for all $y \in Y$, $y \geq x$.

Let X be a partially ordered set, and let $Y \subset X$ be a subset.

- An upper bound $x \in X$ for Y is the *least upper bound* or *supremum* for Y if it minimal among upper bounds for Y, i.e. for any upper bound $x' \in X$ of Y, $x' \leq x$.

  We write $x = \sup(Y)$ (usually pronounced "soup").

- A lower bound $x \in X$ for Y is the *greatest lower bound* or *infimum* for Y if it is maximal among lower bounds for X, i.e. for any lower bound $x' \in X$ of Y, $x' \geq x$.

  We write $x = \inf(Y)$.

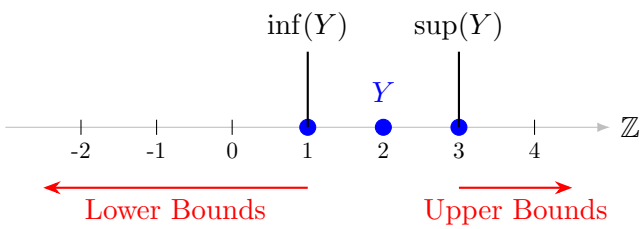


Figure 4.1: $X = \mathbb{Z}$ and $Y = \{1, 2, 3\}$.

Proposition 6. *A subset $Y \subset X$ has at most one supremum and at most one infimum.*

Proof. Suppose $x, x' \in X$ are two suprema[1] of Y. Since any supremum is also an upper bound, we have $x \geq x'$ and $x' \geq x$. By antisymmetry of $\geq$, we have $x = x'$.

The same argument holds for infima. □

Exercise 41. *Show that $\mathbb{Z}$ has no maximum or minimum.*

The set $\mathbb{Q}^+$ of (nonzero) positive rational numbers has no minimum, as the following proposition shows.

Proposition 7. *Let $q > 0$ be a positive rational number. Then there exists a rational number r such that $0 < r < q$.*

Proof. Pick $r = q/2$. We have

$$q - r = q - q/2 = q/2 > 0.$$

□

Note that the supremum and infimum of Y, when they exist, are not necessarily elements of Y. Consider the subset $\mathbb{Q}^+ \subset \mathbb{Q}$.

Proposition 8.

$$\inf(\mathbb{Q}^+) = 0.$$

Proof. Exercise. □

[1] We write suprema instead of supremums because supremum is a Latin word, and we won't let anyone forget it.

4.5 Completeness

Definition 19. *A partially ordered set X is* complete *if every subset $Y \subset X$ with an upper bound has a supremum, and every set $Y \subset X$ with a lower bound has an infimum.*

Example 16. *Any finite ordered set is complete.*

Proposition 9. *The ordered set $(\mathbb{Q}, \leq)$ of rational numbers is not complete.*

Proof. We saw in the previous chapter that there is no rational solution to the equation $p^2 = 2$. Consider the set

$$S = \{x \in \mathbb{Q} \mid x^2 < 2\}.$$

We claim that S has an upper bound, but does not have a supremum. The next three exercises conclude the proof. □

Exercise 42. *Show that every s such that $s^2 > 2$ is an upper bound for S.*

Exercise 43. *Suppose $s^2 < 2$. Then there exists $r \in \mathbb{Q}$, $s < r$ such that $r^2 < 2$. Hence s is not an upper bound for S.*

Hint: Write $M = 2 - s^2$. It is enough to show that there exists $\epsilon > 0$ such that $\epsilon(2s + \epsilon) < M$.

By Proposition 7, there exists $0 < \epsilon < 2s$. In this case,

$$\epsilon(2s + \epsilon) < \epsilon(4s).$$

Thus it is enough to find ϵ such that $\epsilon < 2s$ and

$$\epsilon < \frac{M}{4s}.$$

A second application of Proposition 7 finishes the proof.

Exercise 44. *Suppose $s^2 > 2$. Then there exists $r \in \mathbb{Q}$, $r < s$ which is an upper bound for S. Hence s is not a supremum of S.*

4.6 Axioms for the real numbers

When we defined the natural numbers, we started from a collection of axioms (the Peano axioms) which tell us what we want from the naturals. Only later did we construct a set satisfying these axioms. We now do something similar for the real numbers.

Definition 20 (Formal definition)**.** *We say a set R is a model of the real numbers if it is equipped with:*

- *Elements* $0, 1 \in R$,
- *Binary operations* $+, \cdot : R \times R \to R$,
- *A total order* $\leq$,

such that

- $(R, 0, 1, +, \cdot, \leq)$ *is an ordered field.*
- *The ordered set* $(R, \leq)$ *is complete.*

4.7 A first attempt at constructing the reals

Now that we know what properties we want the reals to have, we can begin to construct them. We will use the informal definition in terms of decimal expansions as a guide. Suppose we are given a real number as a decimal expansion:

$$r = 10.234890234809234....$$

We can *approximate* r with better and better accuracy by consider more and more digits in the expansion:

$$10 < r < 11$$
$$10.2 < r < 10.3$$
$$10.23 < r < 10.24$$
$$...$$

Our basic strategy will be to turn these approximations into a definition. We will think of r as dividing the rational numbers into two camps:

$$A = \{q \in \mathbb{Q} \mid q < r\}$$

and

$$B = \{q \in \mathbb{Q} \mid q > r\}.$$

The key idea is to define r as the unique solution to

$$A < r < B.$$

We will see how to do this in the next section.

4.8 Defining the reals using Dedekind cuts

Definition 21. *A* Dedekind cut *is a pair* (A, B) *of nonempty subsets of* $\mathbb{Q}$ *such that:*

- $\mathbb{Q} = A \cup B$
- *For all* $a \in A$ *and* $b \in B$, $a < b$.
- A *does not have supremum in* $\mathbb{Q}$.

Definition 22. *(Equivalent definition of Dedekind cuts) A subset* $A \subset \mathbb{Q}$ *is called a Dedekind cut, if*

1. $A \neq \emptyset$, $A \neq \mathbb{Q}$,
2. *For every* $x \in A$ *and every* $y \in \mathbb{Q}$ *such that* $y < x$, *we have* $y \in A$,
3. *For every* $x \in A$ *there exists* $y \in \mathbb{Q}$ *such that* $x < y$ *and* $y \in A$.

Exercise 45. *Let* (A, B) *is a Dedekind cut in the sense of Definition 21. Prove that* A *is a Dedekind cut in the sense of Definition 22.*

Conversely, let A *be a Dedekind cut in the sense of Definition 22. Prove that* $(A, \mathbb{Q} \setminus A)$ *is a Dedekind cut in the sense of Definition 21.*

Definition 23. *Let* $\mathbb{R}$ *be the set of Dedekind cuts.*

When we write $(A, B) \in \mathbb{R}$ or say that (A, B) is a Dedekind cut, we mean it in the sense of Definition 21. On the other hand, if only one set A is explicitly mentioned, we imply Definition 22.

We now need to define the structures $0, 1, +, \times, \leq$ and show they satisfy the requisite properties.

First, however, we will show that our definition of the real numbers contains a copy of the rationals.

Definition 24. *We say a Dedekind cut* (A, B) *is* rational *if* B *has a minimum.*

Proposition 10. *The function*

$$(A, B) \to \min(B)$$

is a bijection between rational Dedekind cuts and rational numbers.

Proof. □

We will denote a rational Dedekind cut (A, B) by the rational number $\min(B) \in \mathbb{Q}$ whenever convenient. In particular, we have the distinguished elements

$$0, 1 \in \mathbb{R}$$

corresponding to $0, 1 \in \mathbb{Q}$.

For a rational number $q \in \mathbb{Q}$, we define $q^{\mathbb{R}}$ to be the Dedekind cut $\{x \in \mathbb{Q} : x < q\}$.

Exercise 46. *Show that for each $q \in \mathbb{Q}$, the set $q^{\mathbb{R}}$ is indeed a Dedekind cut. Moreover, show that the function $i : \mathbb{Q} \to \mathbb{R}$ such that $i(q) = q^{\mathbb{R}}$ is an injection.*

Remark 6. *For a rational number $q \in \mathbb{Q}$, show that the set $\{x \in \mathbb{Q} : x \leq q\}$ satisfies the conditions 1 and 2 of Definition 22, but not 3. This explains the importance of the condition 3 – it helps to avoid ambiguity of embedding of $\mathbb{Q}$ into $\mathbb{R}$.*

Next, let's construct the operations of addition and multiplication on Dedekind cuts. Although the following constructions may seem a little surprising or ad-hoc at first, once you play around with them a bit you will find they are quite natural.

Definition 25. *Given Dedekind cuts A and A', define*

$$A + A' = \{a + a' \mid a \in A, a' \in A'\} \tag{4.1}$$

Proposition 11. *The right-hand side of (4.1) is a Dedekind cut. If A, A' are rational, then the sum defined as above coincides with the usual addition of rational numbers.*

Proof. Let us check that $A + A'$ satisfies the conditions 1,2,3 from Definition 22.

1. By Definition 22.1 applied to A and A', we have $A \neq \emptyset$ and $A' \neq \emptyset$. Therefore, we can find elements $a \in A$ and $a' \in A'$. This gives an element $a + a'$ of $A + A'$. This shows that $A + A' \neq \emptyset$.

   By Definition 22.1 applied to A and A', we have $A \neq \mathbb{Q}$ and $A' \neq \mathbb{Q}$. Therefore, we can find elements $b \in \mathbb{Q} \setminus A$ and $b' \in \mathbb{Q} \setminus A'$. Let us show that $b + b' \notin A + A'$. To seek a contradiction, let $b + b' = a + a'$, for some $a \in A$, $a' \in A'$. By Definition 22.2 applied to A, we obtain $\neg(b < a)$, i.e. $a \leq b$. Additionally, since $a \in A$, $b \notin A$, we have $a \neq b$. Therefore, $a < b$. Same argument applied to primed version A' shows $a' < b'$. Thus we have $a + a' < b + a' < b + b'$, which contradicts the earlier established equality $a + a' = b + b'$. Thus we have proved $b + b' \notin A + A'$, and so $A + A' \neq \mathbb{Q}$.

2. Let $x \in A + A'$, and $y \in \mathbb{Q}$ be such that $y < x$. We need to show that $y \in A + A'$.

   By definition of $A + A'$, we have $x = a + a'$ for some $a \in A$, $a' \in A'$. By Definition 22.2 applied to A, we obtain that $y - a' < a$, $a \in A$ imply that $y - a' \in A$. This in turn implies that $y \in A + A'$, as desired.

3. Let $x \in A + A'$. Let present $y \in A + A'$ such that $x < y$.

   Let $x = a + a'$, for some $a \in A$, $a' \in A'$. By Definition 22.3 applied to A and A', we can find $b \in A$ with $a < b$ and $b' \in A'$ with $a' < b'$. Then $y = b + b'$ is the desired element of $A + A'$. Indeed, we have

$$x = a + a' < b + a' < b + b' = y.$$

We leave it as an exercise to check that $A \in \mathbb{Q}$, $A' \in \mathbb{Q}$ imply $A + A' \in \mathbb{Q}$. □

Definition 26. *We define a partial order on $\mathbb{R}$ by*

$$(A, B) \leq (A', B')$$

if $A \cap B' = \emptyset$. Equivalently, in terms of Definition 22, we define $A \leq A'$ if A is a subset of A'.

Proposition 12. *The above relation $\leq$ is (as-claimed) a partial order. It extends the previously defined partial order on $\mathbb{Q}$.*

Proposition 13. *The reals are totally ordered.*

Proof. Exercise. □

For a Dedekind cut $A \in \mathbb{R}$, we define $-A$ to be the unique Dedekind cut such that $(-A) + A = 0^{\mathbb{R}}$.

Exercise 47. *If $A = q^{\mathbb{R}}$ for some $q \in \mathbb{Q}$, then $-A = (-q)^{\mathbb{R}}$.*
If $A \in \mathbb{R}$ is not rational, then $-A = \mathbb{Q} \setminus \{x \in \mathbb{Q} \mid -x \in A\}$.

Definition 27. *Let A, A' be two Dedekind cuts.*
If $0^{\mathbb{R}} \leq A$ and $0^{\mathbb{R}} \leq A'$, we define

$$A \cdot A' = \{x \in \mathbb{Q} \mid \forall b \in \mathbb{Q} \setminus A, \forall b' \in \mathbb{Q} \setminus A' : x < b \cdot b'\}.$$

If $0^{\mathbb{R}} \geq A$ and $0^{\mathbb{R}} \leq A'$, we define

$$A \cdot A' = A' \cdot A = -((-A) \cdot A').$$

If $0^{\mathbb{R}} \geq A$ and $0^{\mathbb{R}} \geq A'$, we define

$$A \cdot A' = (-A) \cdot (-A').$$

Proposition 14. *The operations $(+, \cdot)$ make $\mathbb{R}$ into a field. They restrict to the previously defined $(+, \cdot)$ operations on $\mathbb{Q} \subset \mathbb{R}$.*

Proof. There are many things to check here! I recommend you try verifying one or two of the properties to get a general idea of how the proofs go. Note, however, that essentially all of the proofs piggy-back on the corresponding properties of the rational numbers. □

Proposition 15. *The reals are an ordered field.*

Proof. Exercise. □

Proposition 16. *The reals are complete.*